

Bioinformatics Sequence Alignment Management System

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Bioinformatics Sequence Alignment Management System (BSAMS)** is a computational tool designed to perform optimal global sequence alignment for DNA and protein sequences using the Needleman–Wunsch dynamic programming algorithm. Addressing the fundamental challenge of comparative genomics, BSAMS efficiently calculates alignment scores, accounts for customizable gap penalties, and generates visual representations of matched regions. By integrating robust scoring matrices—such as PAM and BLOSUM for proteins and identity matrices for nucleotides—the system ensures high accuracy in evolutionary and structural analyses. This application demonstrates the practical implementation of dynamic programming in computational biology, providing researchers and students with a reliable framework for sequence homology analysis.

Signature of the Guide